Mark schemes

Q1.

(a) (i) B(50, 0.2)

Use of in (a)

M1

P($R \le 15$) = 0.969 to 0.97

AWFW *0.9692*

A1

2

(ii) P(R = 10) = P($R \le 10$) – P($R \le 9$)

Stated or implied

or

M1

P(R = 10) = $\binom{50}{10}(0.2)^{10}(0.8)^{40}$

Stated or implied

= 0.5836 – 0.4437 = 0.139 to 0.141

AWFW *0.1399*

A1

2

(iii) P(5 < R < 15) = *F*($R \le$ 14 or 15) = 0.9393 or 0.9692

Accept values to 3 dp

M1

minus P($R \le$ 5 or 4) = 0.0480 or 0.0185

Accept values to 3 dp

M1

= 0.89 to 0.893

AWFW *0.8913*

A1

or

B(50, 0.2) expressions stated for at least 3 of 5 < R< 15

Or implied by a correct answer

M1

Answer

(A2)

3

(b) Mean, $\mu = np = 50 \times 0.2 = 10$

Either; CAO

B1

or

Estimate of p, $\hat{p}$ = 0.21

Variance, $\sigma^2 = np(1 - p) = 10 \times 0.8 = 8$

CAO

B1

Mean or **Estimate of *p*** is similar to that expected

10.5 and 10 or 0.21 and 0.2 Either point

but

Variance (standard deviation) is **different** from that expected

20.41 and 8 or 4.5 and 2.8

B1

Reason to **doubt validity** of Sly's claim

Must be based on both 10 or 0.2 and 8 or on both 10 or 0.2 and 2.8 correctly

B1

4

[11]

Q2.

(a) B(15, 0.3)

use of in (a)

M1

(i) (Fewer than) half $\Rightarrow$ 7 or 7½ or 8

stated or implied

B1

Thus require P(K ≤ 7 or < 8)

used or implied by correct answer

M1

= 0.9495 to 0.9505

AWFW (0.9500)

A1

4

(ii) P(2 < K < 7) = 0.8689 or 0.9500

M1

minus 0.1268 or 0.2969

M1

= 0.7415 to 0.7425

AWFW (0.7421)

A1

or

B(15, 0.3) expressions stated for at least 3 terms within 2 ≤ K ≤ 7

or implied by a correct answer

M1

Answer

A2

3

(b) (i) Mean, $\mu = np$ = 15 × 0.4 = 6

CAO

B1

Variance, $\sigma^2 = np(1-p)$ = 6 × 0.6 = 3.6

use of $\sigma^2 = np(1-p)$

M1

Standard deviation = $\sqrt{3.6}$ = 1.89 to 1.9

AWFW; or equivalent

A1

3

(ii) Mean, $\bar{x}$ = 6

CAO (Σx = 60)

CSO if evidence of np(1–p) or 1.9

B1

Standard deviation, s or σ = 2.82 to 2.99

AWFW; or equivalent. ($\Sigma x^2 = 440$)

B1

2

(iii) Means are same/equal

ft on 2 means; accept $\frac{6}{15} = 0.4$

if not contradicted by $\bar{x}$ *in (ii)*

B1ft

Standard deviations are different

dependent on 2 correct SDs

B1 dep

Reason to doubt validity of Kirk's claim

dependent on 2 correct SDs

B1 dep

3

[15]

Q3.

(a) B(15, 0.3)

use of in (a)

M1

(i) P(K = 5) = P(K ≤ 5) – P(K ≤ 4)

P(K = 5) = $\binom{15}{5}(0.3)^5(0.7)^{10}$

may be implied

M1

= 0.7216 – 0.5155 = 0.2055 to 0.2065

AWFW (0.2061)

A1

3

(ii) (Fewer than) half $\Rightarrow$ 7 or 7½ or 8
stated or implied

B1

Thus require P(K ≤ 7 or < 8)
used or implied by correct answer

M1

= 0.9495 to 0.9505
AWFW *(0.9500)*

A1

3

(iii) P(2 < K < 7) = 0.8689 or 0.9500

M1

minus 0.1268 or 0.2969

M1

= 0.7415 to 0.7425
AWFW *(0.7421)*

A1

3

or
B(15, 0.3) expressions stated for at least 3 terms within 2 ≤ K ≤ 7
or implied by a correct answer

M1

Answer

A2

(b) (i) Mean, $\mu = np$ = 15 × 0.4 = 6
CAO

B1

Variance, $\sigma^2 = np(1-p)$ = 6 × 0.6 = 3.6
use of $\sigma^2 = np(1-p)$

M1

Standard deviation = $\sqrt{3.6}$ = 1.89 to 1.9

AWFW; or equivalent

A1

3

(ii) Mean, $\bar{x}$ = 6

CAO *(Σx = 60)*

CSO if evidence of np(1–p) or 1.9

B1

Standard deviation, s or σ = 2.82 to 2.99

AWFW; or equivalent. (Σx^2 = 440)

B1

2

(iii) Means are same/equal

ft on 2 means; accept $\frac{6}{15} = 0.4$ *if not contradicted by* $\bar{x}$ *in (ii)*

B1ft

Standard deviations are different

dependent on 2 correct SDs

B1 dep

Reason to doubt validity of Kirk's claim

dependent on 2 correct SDs

B1 dep

3

[17]

Q4.

(a) (i) p = 0.5

Attempted use of B(14, 0.5) in (a)(i) or (ii)

M1

P($X \leq$ 10) = 0.971 to 0.972

AWFW (0.9713)

B1

(ii) P(X > 5 and X < 10) = P(6 ≤ X ≤ 9)

= P($X \leq$ 9)

Identification of ***at least*** *6, 7, 8 and 9*

M1

$- P(X \leq 5)$

*Identification of **exactly** 6, 7, 8 and 9*

M1

$= 0.9102 - 0.2120 = 0.698$ to 0.699

AWFW (0.6982)

A1

5

(b) $P(Y = 7) = \binom{n}{7}(0.4)^7(0.6)^{n-7}$ M1

Correct expression for
B(7; n, 0.4) with $n \neq 7$

$= \binom{28}{7}(0.4)^7(0.6)^{21}$

Fully correct expression may be implied

A1

$= 0.0425$ to 0.0427

AWFW (0.042556)

A1

3

(c) Different numbers of days in different months

Accept 'n not fixed' OE

E1

1

[9]

Q5.

(a) (i) $p = 0.4$

Attempted use of B(7, 0.4) in (a)

M1

$P(X \leq 2) = 0.419$ to 0.421

AWFW (0.4199)

B1

(ii) $P(X > 1 \text{ and } X < 5) = P(2 \leq X \leq 4)$

$= P(X \leq 4)$

*Identification of **at least** 2, 3 and 4*

M1

$- \quad P(X \leq 1)$

*Identification of **exactly** 2, 3 and 4*

M1

= 0.9037 – 0.1586 = 0.744 to 0.746

AWFW *(0.7451)*

A1

5

(b) $P(Y = 7) = \binom{n}{7}(0.4)^7(0.6)^{n-7}$

Correct expression for B(7; n, 0.4) with n ≠ 7

M1

$= \binom{28}{7}(0.4)^7(0.6)^{21}$

Fully correct expression may be implied

A1

= 0.0425 to 0.0427

AWFW (0.042556)

A1

3

(c) Mean = np = 2.8

CAO

B1

SD = $\sqrt{np(1-p)} = \sqrt{1.68}$

= 1.29 to 1.31

AWFW

B1

2

(d) (i) Mean = 2.8

CAO $\Sigma fx = 140$

B1

SD = 2.24 to 2.27

AWFW $\Sigma fx^2 = 644$

B2

$s_{n-1}^2 = 5.14$ to 5.15 and $s_{n-1}^2 = 5.04$

Substitution of values into correct formula for variance or SD or
SD = 5.03 to 5.15 AWFW

M1

3

(ii) Means are the same

B1ft

SDs differ greatly

ft on (c) and (d)(i);
ft on (c) and (d)(i)
but must be s with σ or s^2 with σ^2

B1ft

Thus answers do not support Aaron's belief

Dependent on B1 above CAO

B1

3

[16]

Q6.

(a) (i) B(n, 0.07)

Use of in (a)

M1

P(X = 2) = $\binom{17}{2}(0.07)^2(0.93)^{15}$

Fully correct expression
May be implied

A1

= 136 × 0.0049 × 0.33670
= 0.224 to 0.225

AWFW (0.22438)

A1

3

(ii) P ($X \leq 5$) | B(50, 0.07))

Attempted; tables or formula (≥ 3 terms stated) May be implied

M1

= 0.865

AWRT (0.8650)

A1

2

(b) B (50, 0.55)

P ($Y \geq 30$) = P ($Y' \leq 20$)

Change from Y to Y′ Must be clear evidence

M1

with p = 0.45

Stated or implied

A1

= 0.286

AWRT (0.2862)

A1

3

[8]

Q7.

(a) (i) B(n, 0.07)

Use of in (a)

M1

$$P(X = 2) = \binom{17}{2} (0.07)^2 (0.93)^{15}$$

Fully correct expression May be implied

A1

$= 136 \times 0.0049 \times 0.33670$
$= 0.224$ to 0.225
AWFW (0.22438)

A1 **3**

(ii) $P(X \leq 5) \mid B(50, 0.07))$
Attempted; tables or formula (≥ 3 terms stated) May be implied

M1

$= 0.865$
AWRT (0.8650)

A1 **2**

(b) <u>B (50, 0.55)</u>

$P(Y \geq 30) = P(Y' \leq 20)$
Change from Y to Y´ Must be clear evidence

M1

with $p = 0.45$
Stated or implied

A1

$= 0.286$
AWRT (0.2862)

A1 **3**

(c) (i) Estimate of $p = \frac{10}{50} = 0.2$
CAO

B1 **1**

(ii) Estimate of SD $(X) = \sqrt{np(1-p)}$
Use of; accept no √

M1

$= \sqrt{50 \times 0.2 \times 0.8} = \sqrt{8}$

= 2.82 to 2.83

AWFW; accept $\sqrt{8}$

A1

2

(iii) SD(X) less than 6.8

Comparison

or

ft on (c)(ii)

M1ft

V (X) less than 46.24

Must be like with like

Not a reasonable assumption

ft on (c)(ii) and like with like comparison

A1ft

2

[13]

Q8.

(a) $n = 16$ $p = 0.85$

$P(D = d) = \binom{n}{d}(0.85)^d (0.15)^{n-d}$

correct expression for B(n, 0.85)with any values of n and d

M1

$P(D = 12) = \binom{16}{12}(0.85)^{12}(0.15)^4$

fully correct expression; may be implied

A1

= 1820 × 0.14224 × 0.00050625

= 0.130 to 0.132

AWFW; accept 0.13

A1

3

(b) $n = 30$ $p = 0.85$

M0 for normal approximation

$P(21 < D < 28) = P(22 \le D \le 27) =$

$P(4 < D' < 9) = P(3 \le D' \le 8)$

attempt at switching to D´
(working with p = 0.15)

M1

P ($D´ \leq / < 8$) or P ($D´ \leq / < 9$)
less than or equal to 8 or 9 less than 8 or 9
(0.9903)

A1

–P ($D´ \leq / < 2$) or P ($D´ \leq / < 3$)
minus
(less than or equal to 2 or 3)
(less than 2 or 3)
(0.3217)

A1

= 0.9722 – 0.1514 = 0.820 to 0.822
AWFW; accept 0.82

A1

OR
At least 3 terms for B(30, 0.85)
or
At least 3 terms for B(30, 0.15)
attempted; may be implied

(M1)

6 to 8 terms (21 to 28) for B (30, 0.85)
or
6 to 8 terms (2 to 9) for B (30, 0.15)
attempted; may be implied

(M1)

= 0.820 to 0.822
AWFW; accept 0.82

(A2)

4

[7]

Q9.

(a) $Y \sim$ Geo (0.8)

B1

1

(b) P(Y = 5) = $(0.2)^4(0.8)$

M1

= 0.00128 =

A1

2

(c) E(Y) = $\frac{1}{p} = \frac{1}{0.8} = 1.25$

CAO

B1

Var(Y) = $\frac{q}{p^2} = \frac{0.2}{0.64} = \frac{5}{16}$

= 0.3125

CAO

B1

2

[5]

Q10.

(a) (i) Binomial n = 25

Attempted use of in part (a)

M1

$P_G = \frac{88}{400}$ (= 22)

cao; may be implied

B1

P(G = 2) = $\binom{25}{2}(0.22)^2(0.78)^{23}$ =

correct expression for B (25, p)
(0 < p < 1) with x = 2

M1

300 × 0.0484 × 0.0032974 = 0.0478 to 0.048

Awfw (0.0478787)
[watch for $(0.22)^2$ = 0.048(4)]

A1

4

(ii) $P_B = \frac{60}{400}$ (= 0.15)

cao; may be implied by correct answer

B1

P($B \le 3$) = 0.4705 to 0.4715

Awfw (0.4711(213))

B1

2

(iii) $P_R = \frac{160}{400}$ (= 0.4)

cao; may be implied by correct answer or ≥1 correct probability

B1

P($8 \le R \le 12$) = P ($R \le 12$)

use of ≤ 12
M1 for ≥ 1 correct term
M2 for 5 correct terms added

M1

– P($R \le 7$)
= 0.8462 – 0.1536

use of – and ≤ 7

M1

= 0.692 to 0.693

Awfw (0.6926(805))

A1

4

(b) Number of trials/events or sample size or n is not fixed

B0 for n not constant or decreasing, etc

B1

P(success) or P(Y) or p is not constant

accept trials/events are not independent or are dependent

B1

2

[12]

Q11.

(a) $n = 18$ $p = 0.15$

$P(Car = 2) =$

binomial used in (a) or (b)

M1

$\binom{18}{2}(0.15)^2(0.85)^{16}$

correct expression

A1

= 0.255 to 0.256

AWFW *(0.2556)*

A1

3

(b) $n = 50$ $p = 0.15$

P $(5 < Car < 10) =$

P $(Car \le 9)$

–P $(Car \le 5)$

Use of ≤ 9 or (6, 7, 8, 9)

M1

Use of – & ≤ 5 or

M1

(4 correct terms added)

= 0.7911 – 0.2194 = 0.571 to 0.572

AWFW *(0.5717)*

A1

3

(c) $n = 900$ $p = 0.15$

$\mu = 900 \times 0.15 = 135$

CAO

B1

$\sigma^2 = 900 \times 0.15 \times 0.85 = 114$ to 115

114.75 (σ = 10.65 to 10.75 AWFW)

B1

P$(Car \le 150)$ = P$(Car < 150.5)$

+ 0.5

B1

$= P\left(Z < \frac{150.5 - 135}{\sqrt{114.75}}\right)$

standardising (149.5, 150, 150.5) using their μ

& their $\sqrt{\sigma^2}$ or correct values

M1

= P(Z<1.45) = Φ(I.45) = 0.926 to 0.927

AWFW *(0.92647)*

A1

5

(d) *p* not 0.15 (value for cars, not all vehicles)
Vehicles not independent

E1

1

[12]

Q12.

(a) (i) *O : A : B : AB = 48 : 35 : 12 : 5*

SC: 3 or 4 correct p values: only B1

Identification of binomial with *n* = 20,
stated or implied anywhere in part (a)

B1

p = 0.48

CAO

B1

P(O = 10) = $\binom{20}{10}$ $(0.48)^{10}(1 - 0.48)^{20-10}$ =

Use of binomial formula with ${}^{20}C_x$; p, 1 – p; x, 20 – x
(M0 for attempted use of tables or normal approximation)

M1

184756 × 0.00064925 × 0.001445551 =

0.1734 = 0.173 to 0.174

AWFW

A1

4

(ii) $p = 0.05$
CAO

B1

$P(AB < 2) = 0.9245 = 0.924$ to 0.925
AWFW

B1

2

(iii) $p = 0.35$
CAO

B1

$P(A \geq 8) = 1 - P(A \leq 7) =$
Use of 0.7624 seen $\Rightarrow$ *B1*

M1

$1 - 0.6010 = 0.40$
AWRT

A1

3

(b) $P(R > 40) = P(R_N > 40.5) =$
CAO

B1

$= P\left(Z > \frac{40.5 - 48}{\sqrt{42.24}}\right)$
standardisation of 39.5, 40, 40.5, 55.5, 56 or 56.5 using their μ *and their* σ
(M0 for attempt at B (400, 0.12))

M1

$P(Z > -1.15) = P(Z < 1.15) = \Phi(1.15)$
attempted area change

m1

$= 0.874$ to 0.877

A1

4

[13]

Q13.

(a) $n = 20$ $p = 0.11$

Identification of binomial in parts (a) or (b)

stated or implied

B1

$P(A = 4) = (0.11)^4 (0.89)^{20-4} =$

$4845 \times 0.00014641 \times 0.154967134 =$

use of binomial formula with $^{20}C_x$; p, $1 - p$; x, $20 - x$

(M0 for attempted use of tables or normal approximation)

M1

$= 0.110$

AWRT; accept 0.11 only with evidence

A1

3

(b) $n = 50$ $p = 0.15$

Identification of B(50, 0.15)

stated or implied by correct answer or by at least one correct term

B1

$P(B \leq 10) = 0.880(1)$

AWRT; accept 0.88 but only from rounding of 0.880(...)

B1

(c) $n = 120$ $p = 0.3$

mean ($\mu = np$) = $120 \times 0.3 = 36$

B1

and

variance ($\sigma^2 = np(1 - p)$) = $120 \times 0.3 \times 0.7 = 25.2$

CAO; or σ = 5.02 AWRT

B1

$P(30 < C_B < 40) = P(30.5 < C_N < 39.5)$

CAO; both

B1

$$= P\left(\frac{30.5 - 36}{\sqrt{25.2}} < Z < \frac{39.5 - 36}{\sqrt{25.2}}\right)$$

standardising (29.5, 30, 30.5) or (39.5, 40, 40.5) using their μ and their σ (not σ^2)
M0 for B(120, 0.3) = 0.623
M0 for 0.623 stated only

M1

= P(–1.10 to –1.09 < Z < 0.69 to 0.70)
AWFW; either

A1

= $\Phi(z_U) - (1 - \Phi(z_L))$ =
attempt at area difference; OE
(must be one +ve and one –ve)

m1

= 0.75804 – (1 – 0.86433)
= 0.617 to 0.623
AWFW; accept 0.62

A1

7

[12]

Q14.

L ~ N(208.5, 2.5^2)

(a) P(205 < L < 210)

$$= \text{P}\left(\frac{205 - 208.5}{2.5} < Z < \frac{210 - 208.5}{2.5}\right)$$

Standardising (204.5, 205, 205.5 or 209.5, 210, 210.5) with f.t. 2.5, 2.5 or 2.5^2; allow (208.5 – l)

M1

= P(–1.4 < Z < 0.6)
CAO; either ignoring sign

A1

= $\Phi(0.6) - (1 - \Phi(1.4))$
Area difference

m1

= 0.72575 – (1 – 0.91924)

= 0.72575 – 0.8076 = 0.64499 or 0.6448 to 0.6452

AWFW; AG of 0.645

A1

4

(b) Binomial

Use of

M1

P(R = 6)

$= \binom{10}{6}(0.645)^6 = ((1 - 0.645) = 0.355)^4$

Correct expression; may be implied

A1

= 210 × 0.0720 × 0.01588 = 0.239 to 0.241

AWFW

A1

3

[7]

Q15.

n = 50 and p_R = 0.2

(a) Model applicable is Binomial

Use of in (i) or (ii)

M1

P(R < 10) = P($R \leq 9$)

Stated or implied from tables or by formulae

M1

= 0.4437

AWFW 0.443 to 0.444

A1

3

(b) P(8 ≤ R ≤ 12) = P($R \leq 12$)

Stated or implied from tables or by formulae

M1

–P($R \leq 7$)

A1

= 0.8139 – 0.1904 = 0.6235

AWFW 0.623 to 0.624

A1

3

[6]

Q16.

(a) Binomial (n, 0.75)

CAO, attempted use anywhere

B1

P(X = 10) = $\binom{16}{10}(0.75)^{10}(0.25)^{6}$

Use of B(16, 0.75) and 10

M1

= 8008 × 0.75^{10} × 0.25^{6}
= 0.11

AWRT

A1

3

(b) P($Y \geq 20$ | n = 30, p = 0.75)

Attempted switch
Use of normal MO

M1

P($Y' \leq 10$ | n = 30, p = 0.25)

$Y' \leq 10$
P = 0.25

A1/A1

= 0.89 to 0.90

AWFW 1-(0.89 TO 0.90) M1A1A1AO

A1

(b)

	B(30,0.75)		B(30,0.25)
y	P($Y = y$)	y'	P($Y' = y'$)

Use of binomial formula

(M1)

20	0.0909	0	0.0002
21	0.1298	1	0.0018

22	0.1593	2	0.0086

Attempt at 20 to 30
or at 0 to 10

(m1)

23	0.1662	3	0.0269
24	0.1455	4	0.0604
25	0.1047	5	0.1047

correct use of 0.75 or 0.25

(A1 dep on m1)

26	0.0604	6	0.1455
27	0.0269	7	0.1662
28	0.0086	8	0.1593
29	0.0018	9	0.1298
30	0.0002	10	0.0909
Sum	**0.8943**	Sum	**0.8943**

AWFW 0.89 to 0.90

(A1)

4

[7]

Q17.

$n = 20$ $p_0 = 0.4$ $p_R = 0.6$

(a) P(O = 10) or P(R = 10)

identification of B (20, 0.4 OR 0.6) in (a) or (b)

B1

$= \binom{20}{10}(0.4)^{10}(0.6)^{10}$

use of formula with $^{20}C_r$, $p + q = 1$, Σpowers = 20 or tables ($\le$ 10 – $\le$ 9)
(M0 for normal approx)
(0.117 × 2 $\Rightarrow$ B1, M1, AO)

M1

184756×0.000104857 × 0.006046617 = 0.8725 – 0.7553
= 0.117

AWRT

A1

3

(b) $P(O < 10) = P(0 \le 9)$

attempt at $\le$9 (or $\ge$ 11)
(M0 for normal approx)

M1

= 0.755

AWRT

A1

2

(c) Sample size or n is not fixed

OE

n = 0 not available $\Rightarrow$ E1

Geometric $\Rightarrow$ E0

E1

1

[6]

Q18.

(a) $F = 0.12 \quad M = 0.53 \quad S : 0.35$

Identification of binomial with n = 50, stated or implied anywhere in question

B1

$P(F = 6) = \binom{50}{6}(0.12)^6(1 - 0.12)^{50-6} =$

Use of binomial formula with
${}^{50}C_x$; p, $1-p$; $50, 50-x$
(M0 for attempted use of tables or normal approximation)

= 15890700 × 0.000002985 × 0.003607759

0.170 to 0.172

AWFW

A1

3

(b) $P(S \leq 20) =$

identification of ≤20; seen or implied (M0 for attempted use of normal approximation)

M1

0.813 to 0.815

AWFW; tables give 0.8139

A1

2

[5]

Q19.

(a) (i) Binomial n = 50 p = 0.4

Binomial

B1

P(20 or fewer) = 0.5610

n = 50 p = 0.4

B1

0.561 (0.5605 ~ 0.5615)

B1

3

(ii) 20 or fewer invigorating → 30 or more relaxing

reasonable attempt to express in terms of number of relaxing cubes or method for calculating 20 or fewer invigorating

M1

P(30 or more) = 1 – P(29 or fewer)

P(30 or more) = 1 – P(29 or fewer)

m1

1 – 0.9966
= 0.0034

0.0034 (0.00335 ~ 0.00345)

A1

3

(iii) More relaxing → 26 or more relaxing

M1

P(26 or more) = 1 – P(25 or fewer)
= 1 – 0.9427

reasonable attempt to express more relaxing in terms of number of relaxing cubes

= 0.0573

0.0573 (0.057 ~ 0.0574)

A1

2

(b) (i) Not binomial, n not fixed

not binomial

M1

n not fixed

A1

(ii) Not binomial, p not constant/not

not binomial

M1

independent

reason

A1

4

[12]

Q20.

(a) (i) Binomial $n = 8$ $p = 0.3$

Binomial

8, 0.3

B1 B1

P(2 or fewer) = 0.552

0.552 (0.551, 0.5525)

B1

(ii) P(2) = 0.5518 – 0.2553 = 0.2965

P(2 or fewer) – P(1 or fewer) or use of correct formula

M1

0.2965(0.296, 0.297)

A1

(iii) P(>3) = 1 – 0.8059 = 0.194

P(>3) = 1 – P(3 or fewer) or use of correct formula

0.194(0.193, 0.195)

A1

7

sc *B1 0.448 (0.448 , 0.449)*

(b) No, n not constant/probabilities not random/not independent/ 0,1 not possible outcomes

No

Reason

M1 A1

2

(c) No, p not constant/ not independent

No

Reason

M1 A1

2

[11]

Q21.

(a) (i) Binomial $n = 5$ $p = 0.15$

Binomial

B1

$n = 5$ $p = 0.15$

B1

p (2 or fewer) = 0.973

0.973 (0.973 to 0.974)

B1

3

(ii) p (more than 3) = 1 – 0.9978

$p(>3) = 1 - P$ (3 or fewer)

M1

= 0.0022

0.0022(0.00215 to 0.002250) Allow 0.002

A1

2

(iii) mean = 5 × 0.15 = 0.75

0.75 ***cao***

B1

s.d = $\sqrt{5}$ × 0.15 × 0.85 = $\sqrt{0.6375}$

Method – allow variance if called variance even if incorrect late variance = 0.6375

M1

= 0.798

0.798 (0.798 to 0.8)

sc *Use of n = 7 in this question*

(a)(i) Binomial n = 7 p = 0.15

0.9262 3

(ii) 0.0121 2 MR – 1

(iii) 1.05 0.945   3
Total 7

No further allowance in (b) and (c) (if they continue with n = 7 this would be a second misread and only (b)(ii) is affected).

A1

3

(b) (i) mean = 0.75

0.75 ***cao***

B1

s.d = 1.24 or 1.23

1.24 (1.23 to 1.2) allow M1A1 if method shown

B2

3

(ii) Proportion = $\frac{0.75}{5}$ = 0.15

Their mean divided by 5

E1f.t.

1

(c) (i) Not plausible
s.d. much larger than for binomial

B1

s.d. too large – allow arguments based on probabilities calculated in (a)(i) and (ii)

E1f.t.

2

(ii) *p* not constant – some pupils more likely to be late than others

p not constant / not independent

E1

late arrivals not independent – may be due to weather/transport difficulties etc

For 2 marks, need 2 reasons with at least one in context.

E1

2

[16]

Q22.

(a) (i) Binomial n = 6 p = 0.5

B(6, 0.5)

B1

P(more than 4) = 1 – 0.8906

P(more than 4) = 1 – P(4 or fewer)

M1

= 0.109

0.109 (0.109 to 0.11)

A1

3

(ii) P(6) = 1.0000 – 0.9844 = 0.0156

P(6) = 1 – P(5 or fewer) or
P(6 or fewer) – P(5 or fewer)
or correct use of formula

M1

0.0156 (0.015 to 0.016)

A1

2

(b) 14 out of 900 = 0.0156

Appropriate calculation attempted

M1

It appears the proportion of unit trusts outperforming the stock market average

Conclusion consistent with their earlier results

E1f.t.

over a six-year period is consistent with a random selection of investments

Appropriate conclusion based on correct calculations

E1

3

[8]

Q23.

(a) (i) $z = \frac{75 - 85}{8} = -1.25$

Method for z; ignore sign

M1

P(< 75) = 1 – 0.89435

A correct use of normal tables

M1

= 0.106

0.106 (0.105 to 0.106)

A1

3

(ii) $z_2 = \frac{81-85}{8} = -0.5$

Completely correct method; allow both z's positive

M1

Probability between 75 and 85 is

0.89435 – 0.69146 = 0.203

Reasonable attempt, both z's negative

M1

0.203 (0.202 to 0.204)

A1

3

(b) 85 + 3.0902 × 8 = 110

3.09 or 3.0902

B1

(their z) × 8

M1

Completely correct method

m1

110 (109 to 110)

A1

4

(c) $z = \frac{81-85}{\frac{8}{\sqrt{4}}} = -1$

Use of $\frac{8}{\sqrt{4}}$

M1

Correct method for z

m1

Probability mean less than 81 = 1 – 0.84134 = 0.159

Completely correct method

m1

0.159 (0.158 to 0.16)

A1

4

[16]

Q24.

(a) (i) Binomial n = 20 p = 0.15

Binomial n = 20 p = 0.15

B1

P(0) = 0.0388

0.0388 (0.0387 – 0.039)

B1

(ii) P(1 or more) = 1 - 0.0388 = 0.961

M1

P(1 or more) = 1 – P(0)

A1

4

0.961 (0.961 – 0.962)

(b) (i) Binomial

B1

n = 50 p = 0.08

B(50, 0.08)

B1

P(2 or more) = 1 – 0.0827

P (2 or more) = 1 – P (1 or fewer) - may be earned in (ii)

M1

= 0.9173

(ii) P (3 or more) = 1 – 0.2260 = 0.774

0.917 (0.917 – 0.918) and
0.774 (0.7735 – 0.7745)

A1

4

(c) P(1 or more exec) × P(2 or more stand)

Reasonable attempt to define ways of seats being available

M1

Correct ways of seats being available

m1

+ P (0 exec) × P(3 or more stand)

Method for probability, depends on previous M mark only in (c)

M1

Completely correct method (dependent on Ms in (a) and (b) and (c)

m1

0.9612 × 0.9173 + 0.0388 × 0.774 = 0.912

0.912 (0.91 - 0.914)

A1

5

[13]

Q25.

(a) Binomial n = 40 p = 0.35

Binomial

n = 40, p = 0.35

B1 B1

P(9 or fewer) = 0.0644

0.0644(0.064 – 0.065)

B1

3

(b) P(7 or more) = 1 – 0.0044

P(7 or more) = 1 – P(6 or fewer) generous, or correct formula

Allow 1 – P(7 or fewer) = 1 – 0.0124

M1

= 0.9956

0.9956 (0.995 – 0.966)

A1

2

(c) P(20) = 0.9827 – 0.9637 = 0.0190

P(20) required

P(20) = P(20 or fewer) – P(19 or fewer) or correct formula – can be earned for P(r) $r \neq 20$

0.0190 (0.0189 – 0.0191)

B1 M1 **3**

[8]

Q26.

(a) $\bar{x}$ = 12 s = 1.73

12 cao

1.73(1.73 – 1.735) allow

1.63(1.63 – 1.635)

0.4 cao

B1/B1 **2**

(b) 0.4

0.4 cao

B1 **1**

(c) Binomial

B1 **1**

(d) (i) P(X > 15) =0.9029

attempted use of B(30,p), their p method – allow P(≥15)

B1/M1

= 0.0971

0.0971(0.097 – 0.0972)

A1 **3**

(ii) Mean 12

12 cao

B1

s.d. = $\sqrt{30 \times 0.4 \times 0.6} = 2.68$

Method – allow variance if called variance, their p

2.68(2.68 – 2.69)

M1/A1

3

(e) Unsuitable

Unsuitable

E1

standard deviations very different

ft s.d. different

max 1 if previous M not earned

E1 ft

2

(Mean same but this would be the case no matter what the distribution of X)

(f) P not constant – some houses more likely to put boxes out than others

P not constant/not independent

E1

1

[13]

Q27.

(a) B(40,0.09)

binomial, n=40 p=0.09

any correct use of binomial tables or formula;

allow small slip eg. reading p = 0.08

B1 M1

P(2 or fewer) = 0.289

0.289 – 0.290

A1

3

(b) P(4) = 0.7103 – 0.5092

M1

= 0.201

0.2 – 0.202

A1

2

(c) P(more than 5) = 1 – 0.8535

any correct method

allow 1 – P(4 or fewer) – implied by 0.2897

M1

= 0.1465

0.146 – 0.147

A1

2

[7]

Q28.

(a) (i) Binomial n = 10 p = 0.4

B1

P(3 or fewer) = 0.382

Correct use of tables ***or*** *correct method of calculation*

0.382-0.383

M1 A1

3

(ii) (> 4) = 1 – (4 or fewer) = 1 – 0.6331

(> 4) = 1 – P(4 or fewer) ***or***
correct method of calculation

M1

= 0.367

0.366 – 0.368

A1

2

(b) Mean = 10 × 0.4 = 4

cao

B1

Standard deviation $= \sqrt{10 \times 0.4 \times 0.6} =$

Method for s.d. or variance (if called variance)

M1

1.55

1.54 – 1.56

A1

3

(c) 3.92

3.92 – 3.93

B1

3.17 (allow 3.05)

3.17 – 3.18 (3.045 – 3.055)

B1

2

(d) Mean similar but standard deviation much greater than model

Standard deviations different

E1

Binomial not good model

Binomial not suitable model

E1

Siballi's beliefs not plausible

Siballi's beliefs not supported

E1

3

[13]

Q29.

(a) (i) Binomial

B1

n =40 p = 0.04

B1

P(4 or fewer) = 0.979

B1

0.979(0.9785 – 0.9795)

(ii) P(2) = 0.7855 – 0.5210 = 0.2645

P(2) = P(2 or fewer) – P(1 or fewer)

Completely correct method

0.2645(0.264 – 0.265)

M1 m1 A1

6

(b) 40 completed $\rightarrow$ 0 not completed

0 not completed

M1

$P(0) = 0.195$

A1

2

0.195(0.195 – 0.196)

[8]